

Etiopathology, clinical features, and treatment of diffuse and multinodular nontoxic goiters

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Abstract Goiter, an enlargement of the thyroid gland, is a common problem in clinical practice associated with iodine deficiency, increase in serum thyroid-stimulating hormone (TSH) level, natural goitrogens, smoking, and lack of selenium and iron. Evidence suggests that heredity also has an important role in the etiology of goiter. The current classification divides goiter into diffuse and nodular, which may be further subdivided into toxic (associated with symptoms of hyperthyroidism, suppressed TSH or both), or nontoxic (associated with a normal TSH level). Nodular thyroid disease with the presence of single or multiple nodules requires evaluation due to the risk of malignancy, toxicity, and local compressive symptoms. Measurement of TSH, accurate imaging with high-resolution ultrasonography or computed tomography, and fine-needle aspiration biopsy are the appropriate methods for evaluation and management of goiter. This review discusses the clinical presentation, diagnostic evaluation, and treatment considerations of nontoxic diffuse and nodular goiters.

Keywords Multinodular goiter · Physiopathology · Clinical evaluation · Treatment

Introduction

Goiter, a common problem in clinical practice, is characterized by a diffuse or localized enlargement of the thyroid at any stage of life. It is associated with changes in the thyroid's ability to

secrete adequate amounts of hormones as a result of intrinsic glandular defects or exogenous factors that modify the normal activity of the gland. Depending on the elements involved in this process and the efficacy of the adaptive mechanisms, the enlargement of the gland often prevents the development of hypothyroidism. Regardless of clinical presentation, the development of a goiter is essentially the same: an attempt of the thyroid to adapt to circumstances of impaired hormone secretion.

According to morphological characteristics, a goiter may be classified as diffuse or nodular. The term “nodular thyroid disease,” which encompasses single and multiple palpable or nonpalpable nodules, is more descriptive and accurate than the currently used terms “solitary thyroid nodule” and “multinodular goiter.” In an epidemiological context, the terms sporadic or endemic are used. Endemic goiter refers to a circumstance in which goiter is present in more than 10 % of the population due to chronic iodine deficiency. A nodular goiter may be either toxic or nontoxic. In contrast to solitary nodular disease, which displays more uniform clinical, pathological, and molecular pictures, multinodular goiter belongs to a group of mixed nodular entities. The same gland often presents a combination of thyroid lesions producing excessive, deficient, and normal amounts of hormones. The balance of the functional properties of individual nodules present in a nodular goiter determines the functional status of the thyroid, which may be of normal function, or subclinical or overt hyperthyroidism. In the absence of thyroid dysfunction, inflammation, or malignancy, nodular goiter along with diffuse goiter is a clinical entity identified as simple or nontoxic goiter.

Nontoxic diffuse goiter (NDG)

An NDG is an enlargement of the thyroid that occurs regardless of circulating TSH levels and affects individuals living in areas of iodine sufficiency (sporadic NDG)

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or, more frequently, of mild to moderate iodine deficiency (endemic NDG). In addition to the higher level of iodine intake, sporadic NDG, when compared with endemic NDG, develops in younger individuals who are usually transitioning from puberty into adulthood, grows slowly but continuously, and is associated with normal thyroid function tests (serum thyroid-stimulating hormone [TSH] and free thyroxine [free T4] [1].

A persistent increase in TSH stimulation due to chronic iodine deficiency in NDGs leads initially to a homogeneous increase in thyroid size, but maintenance of the normal morphology of the follicles. The structural heterogeneity that develops later in NDGs may be due to a greater sensitivity of different follicles to TSH stimulation, even when TSH is within normal levels [1].

Strong arguments support the hypothesis that both sporadic and endemic NDGs result from an interaction of the events mentioned above. Iodine deficiency is a well-recognized dominant (and probably permissive) factor in this setting, particularly in endemic goiters. Other factors, such as dietary (Table 1) and environmental chemicals (Table 2), as well as heterogeneous genetic abnormalities of the thyroid may contribute to the development of NDG, although the influence of these factors seems to only be substantial in conditions of low iodine intake [2].

Table 1 Natural goitrogenic agents and their mechanisms of action

	Mechanism
Foodstuffs	
Lima beans, linseed, sorghum, sweet potato, cassava	Contain cyanogenic glycosides; metabolized to thiocyanates that compete with iodine in the thyroid uptake
Cruciferous vegetables: cabbage, cauliflower, kale, broccoli, radish, rapeseed	Contain glucosinolates; metabolites compete with iodine in the thyroid uptake
Soy beans, millet, babassu coconut, manioc	Contain flavonoids that reduce the activity of thyroperoxidase
Seaweed (kelp)	Iodine excess inhibits release of thyroidal hormones
Nutrients	
Selenium deficiency	Accumulated peroxides can damage the thyroid; deiodinase deficiency alters hormone synthesis
Iron deficiency	Reduces the activity of heme-dependent thyroid thyroperoxidase and may reduce the effectiveness of iodine prophylaxis
Vitamin A deficiency	Increases thyroid size by increasing pituitary TSHB mRNA and TSH

TSHB thyroid-stimulating hormone beta subunit, *mRNA* messenger RNA, *TSH* thyroid-stimulating hormone. Modified and adapted from [8]

Nontoxic nodular goiter (NNG)

An NNG is characterized by a nodular increase of the thyroid resulting from multifocal monoclonal or polyclonal proliferation of thyrocytes producing groups of new follicles or structures similar to follicles considerably heterogeneous from functional and morphological points of view.

The prevalence of NNG in a population correlates inversely with the iodine intake [3]. Based on ultrasonographic studies in iodine-deficient areas, the reported frequency of nodular disease in adults is about 30–40 % in women and 20–30 % in men. Small differences in the environmental supply of iodine may explain the diverse rates of thyroid abnormalities found in various populations. Differences reported in association with iodine intake include goiter frequency (15 % with a mild and 23 % with a moderate deficiency), and nodular size (increased in individuals with moderate iodine deficiency) [4].

The frequency of nodules seems to increase with age. In an area of borderline iodine deficiency in Denmark, NNG was found in 23 % of 2656 individuals aged 41–71 years and increased with age [5].

On the other hand, the association between age and thyroid volume is conflicting. In areas of iodine deficiency (except for those with severe deficiency) and sufficiency, the volume of the thyroid is reported to peak and plateau around the age of 40 years. In contrast, in the iodine-sufficient Whickham cohort, the frequency of goiter decreased with age along 20 years of follow-up both in women (from 23 to 10 %) and men (from 5 to 2 %) [6].

Etiology of NDG and NNG

Nodular thyroid disease is a heterogeneous disorder divided into solitary nodular and multinodular. Histologically, benign thyroid nodules may be divided into true adenomas (encapsulated lesions) or adenomatous nodules (non-encapsulated lesions). These nodules may also be divided according to morphological criteria following the World Health Organization (WHO) classification [7]. In contrast to solitary nodular disease which has a more uniform clinical, pathologic, and molecular picture, NNG is a mixed nodular entity.

The same primary and secondary factors implicated in the etiology of NDGs mentioned above also exert potential influence on the development of NNGs [1] (Fig. 1). However, in NNGs, these factors culminate after decades with development of areas of necrosis, hemorrhage, marked fibrosis, and calcification [8]. Interestingly, it has been demonstrated that nodular areas within the same gland exhibit different uptakes of iodine and generation

Table 2 Environmental chemical compounds with potential deleterious effects on the human thyroid system

Agents	Class	Mechanism	Effects on thyroid hormones
Perchlorate, thiocyanate, phthalate, chlorate, bromate, disulfide from coal processing, tobacco	Iodide transport	Competition/blocking of SLC5A5 or NIS	Decrease in thyroid synthesis of T3 and T4
Methimazole, propylthiourea, amiloride (herbicide), benzophenone 2 (used as sunscreen in cosmetics), soy isoflavone, mancozeb (fungicide)	Synthesis inhibitors	Inhibition of TPO	Decrease in thyroid synthesis of T3 and T4
PCBs, pentachlorophenol (wood preservative), PBDEs (flame retardants)	Transport deregulation	Competitive binding to transthyretin	Interfere with TH-dependent neural differentiation
Acetochlor (herbicide), PCBs	Increased hepatic catabolism	Stimulation of glucuronyltransferases or sulfotransferases	Increase in biliary excretion of T3 and T4
Dioxin, flame retardants, carbonitrile, bisphenol A (plastic component, inner lining of tin cans for food), PCBs	Increased cellular transport	Stimulation of transporters OATP1C1 or MCT8	Increase in biliary excretion of T3 and T4
PCBs, triclosan (bactericidal), pentachlorophenol, dioxins, difuran	Inhibition of sulfation	Inhibition of sulfotransferases	Decrease in TH sulfation leading to possible reduction of T3 peripheral synthesis
FD&C red dye #3, propylthiouracil, PCBs, octyl-methoxycinnamate (UV blocker)	Deiodinases	Inhibition or stimulation of deiodinases	Decrease in peripheral synthesis of T3
PCBs, bisphenol A, hexachlorobenzene, flame retardants	Agonists or antagonists of the TH receptor (THR)	Direct or indirect interference in THR binding	Alteration of TH-dependent gene transcription
DDT, PCBs	TSH receptor	Inhibition of TSHR	Decrease in T3 and T4 production

Adapted from [21]

PCBs polychlorinated biphenyls, FD&C Red Dye #3: 2',4',5',7'-tetraiodofluorescein disodium salt, or erythrosine, red dye used in food, drugs, and cosmetics, SLC5A5: solute carrier family 5 (sodium-iodide symporter), member 5 or NIS (sodium-iodide symporter), TPO thyroperoxidase, THR TH nuclear receptor, OATP1C1 or SLC01C1 solute carrier organic anion transporter family, member 1C1, MCT8, or SLC16A2 solute carrier family 16, member 2 (monocarboxylic acid transporter 8), TH thyroid hormone, TSHR thyroid-stimulating hormone receptor, PBDE polybrominated diphenyl ether, DDT 1,1,1-trichloro-2,2-bis(4-chlorophenyl)ethane

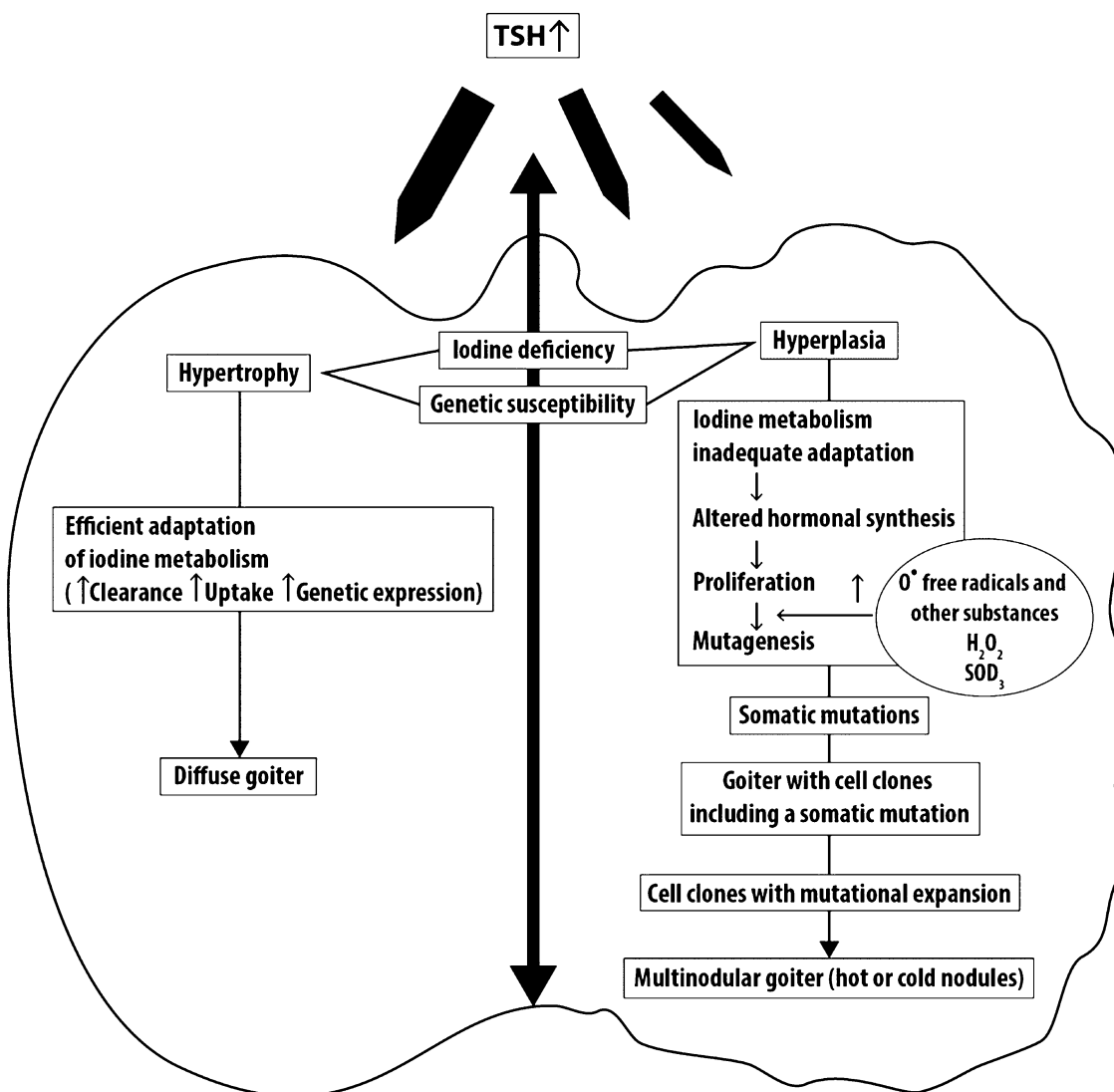


Fig. 1 Hypothetical physiopathology of a multinodular goiter. Hyperplasia induced by a goitrogenic stimulus (i.e., iodine deficiency) is the earliest step in the development of a nontoxic nodular goiter (NNG). The natural heterogeneity among thyrocytes results in thyroid follicles that are very diverse in their growth capacity, as well as their sensitivity and response to TSH. However, NNG growth seems to be essentially TSH-independent, at least in the later stages. In addition to an increase in functional activity, there is a great increase in thyroid cells number. Since the growth of such cells is

periodical, the goiter is prone to develop nodules over time. This thyroid cell replication leads to increased mutagenesis either by a direct (through the production of H_2O_2 /free radicals) or indirect (cell proliferation and increased cell division) mechanism. Subsequently, hyperplasia leads to the development of heterogeneous cell clones. Some of these clones show somatic mutations in the thyroid-stimulating hormone receptor (TSHR), originating nodules with autonomous (“hot”) function or with mutations that lead to dedifferentiation (“cold” nodules or adenomas). Adapted with modifications from Ref. [22]

of cyclic AMP, confirming the functional heterogeneity of NNGs, as discussed below [9].

Until recently, NNG was mainly regarded as a consequence of iodine insufficiency and persistent (although often only marginal) TSH elevation leading to thyrocyte proliferation and diffuse enlargement of the gland during childhood and adolescence [10]. According to this hypothesis, nodules would subsequently appear with age, leading to the development of a multinodular goiter. While the important roles of iodine deficiency and the growth-promoting

contribution of TSH are still recognized, this hypothesis has been revised. An NNG is now considered a consequence of an inherent propensity of the thyroid to develop nodules with age, amplified by the presence of additional factors further promoting thyrocyte proliferation and nodule formation [1].

An analysis based on the Mendelian principles has confirmed the complexity involved in the hereditary transmission in NNG. It became clear more recently, as discussed below, that both sporadic and endemic NNG belong to a

group of complex diseases. Their clinical presentation is influenced by genetic and environmental factors and both vary in severity.

A substantial number of NNGs occur outside areas with scarce iodine supply and in patients who have never been exposed to iodine deficiency. The fundamental process of goitrogenesis is independent of iodine deficiency and operates through mechanisms innate to the hereditary and acquired heterogeneity within the thyrocytes themselves. However, superimposed iodine shortage, even at moderate degrees, greatly enhances the incidence of NNG and shifts its clinical appearance toward younger individuals (school-children and young adults) with the additional influence of enhanced TSH secretion (Fig. 1). This fact has been confirmed in a population study in Denmark, where the iodine consumption *per capita* has been recently described as low [8].

It should be mentioned that the constitutional factor represented by the female gender is involved in the etiology of goiter on a population level, since the ratio of affected women to men in nonendemic regions is higher than 5–1 (some studies estimate a proportion of 10–1), although the reasons for this are poorly understood. Thus, at present, one can only speculate the occurrence of a genetic susceptibility to thyroid disease or a direct impact of steroid hormones. In fact, a growth-promoting effect of estrogen has been described in vitro in rat FRTL-5 cells and thyroid cancer cell lines and has been proposed as a possible contributing and constitutional effect of gender [11]. In addition, it has been suggested that 17 β -estradiol amplifies the growth factor-induced signaling in the normal thyroid and in thyroid tumors [12].

Smoking is another unquestionable environmental factor that contributes to the development of NNGs, especially in areas of mild iodine deficiency. It has been confirmed that the thiocyanate present in tobacco competes with SLC5A5 (solute carrier family 5 [sodium-iodide symporter], member 5, or NIS [sodium-iodide symporter]) for the active uptake of iodine in the basal membrane of the thyrocyte [13, 14].

Other factors associated with an increased risk of development of thyroid nodules and goiter include pregnancy (which is associated with an increased prevalence of thyroid nodules and goiter in iodine-deficient, but not iodine-replete areas) [14], and presence of uterine fibroids (which is associated with a two-fold increased risk of thyroid nodules, probably related to estrogen and other physiopathological mechanism) [15]. The pathogenesis of the association between goiter and oral contraceptives remains to be settled and is probably a combination of several direct and indirect effects of sex steroids on the thyroid [16]. Statins seem to be associated with goiter by an antiproliferative and pro-apoptotic effect of the drug on thyroid cells [17],

whereas the association of alcohol consumption and a lower prevalence of goiter and thyroid nodules is currently unclear [14].

Dietary goitrogenic agents that interfere with iodine uptake by the thyroid (such as cassava thiocyanate) may be considered important in the prevalence of goiter, since these agents block the incorporation of iodide into the follicular cell, similarly to pharmacological agents such as propylthiouracil (PTU) or methimazole. These agents frequently exert a potentiating action in conditions of relatively low dietary iodine [8]. This fact has been clearly observed in some areas in Brazil (in the south of Maranhão and Piauí states and part of the Tocantins state) which have a high prevalence of goiter. The local population consumes large amounts of products derived from babassu coconut (edible oil, pressed coconut, and babassu “flour”), which is present in the inner layer of the seed [18]. It has been documented that the skin of the coconut and the “flour” have a PTU-like action that blocks the enzyme peroxidase with very low efficiency (Table 1). Paradoxically, excessive dietary iodine can also lead to the development of NDG [19]. This has been documented in a study performed in the northern part of the Japanese archipelago (Sapporo, Hokkaido) in which the population maintains high blood iodine concentration due to chronic ingestion of foods rich in iodine such as sea algae. The most accepted hypothesis is that the incorporation of iodine into tyrosine residues of the thyroglobulin (TG) molecule is blocked hindering the synthesis of triiodothyronine (T3) and T4. With this inhibition (known as the Wolff–Chaikoff effect) there is a decrease in circulating thyroid hormones and consequent increase in TSH, which leads to an increase in thyroid volume. In that study, removal of sea algae from the diet overtly reduced goiter in the participating individuals. Certain medications acting on the thyroid eventually result in thyromegaly. For instance, chronic and continuous therapy with lithium in certain psychiatric disorders may inhibit thyroid hormone release, resulting in lower circulating hormone availability. Goiter is a common side effect of lithium. Among individuals with bipolar disorder treated with lithium, up to 40 % develop goiter and about 20 % become hypothyroid [20].

In addition to medications, environmental chemicals can impact thyroid function and incidentally contribute to the development of goiter. Known endocrine disruptors include exogenous agents that interfere with the production, release, transport, metabolism, binding, action, or elimination of natural hormones, particularly thyroid hormones, responsible for the maintenance of homeostasis, regulation of growth, and behavior. Considering only their effects in humans, these substances are foreign to the body, causing deleterious effects to individuals or their descendants as a result of interferences on the endocrine system [21].

Thyroid disruptors can affect the physiology of the gland at several stages, interfering with the hypothalamic-pituitary-thyroid axis, inhibiting the active transport of inorganic iodine to the thyrocyte, mimicking or antagonizing the action of thyroid hormones, inducing the expression or inhibiting regulatory enzymes associated with the synthesis and metabolism of thyroid hormones, and modifying the levels of thyroid hormone receptors or signal transduction resulting from hormonal action [21]. These features are depicted in (Table 2).

Even though the genetic involvement is complex in nature and has many areas which are not yet fully understood, it must be regarded as decisive in the development of NDG and NNG, mainly in circumstances of moderate iodine dietary deficiency [22–27]. Therefore, we must accept that the presence of one or more of the environmental factors listed in Table 1 along with individual genetic determinants will ultimately be responsible for triggering processes that lead to goiter development. The occurrence of NDG during adolescence and associated with a clear familial occurrence, even in conditions of normal dietary iodine intake, is highly indicative of a genetic susceptibility. In contrast with familial NDG caused by random genetic variations due to spontaneous mutations, some families have an autosomal dominant pattern of inheritance [25, 26]. Gene–gene interactions and polygenic mechanisms (i.e., synergistic effects of genetic variations or polymorphisms) indicate the complexity of the pathogenesis of NDG or NNG. In addition, genetic predisposition to the disease has been evidenced in studies with thorough familial genetic screening, as well as studies carried out with twins [23, 24].

Children born to parents with sporadic NDG have a significantly higher risk of developing goiter in adulthood when compared with offspring of unaffected parents [23]. The elevated incidence of NDG in women and its high concordance in monozygotic, when compared with dizygotic twins, reinforce the influence of a genetic predisposition [23]. Another consonant aspect is the recurrence of goiter after partial thyroidectomy, which can occur months or years after surgery. Recurrence seems to be linked to autosomal recessive genetic defects in the thyroid system, suggesting the occurrence of putative dysmorphogenesis in familial euthyroid goiter [22]. Genetic defects similar to those found in hypothyroid patients but with minor functional consequences are likely to be compensated with the development of a euthyroid goiter, especially in association with iodine deficiency [22].

Most of these defects could affect candidate genes involved in the physiology of the thyroid: *TG*, *TPO*, *NIS* (*SLC5A5*), *SLC26A4* [solute carrier family 26 (anion exchanger), member 4], *PDS* (pendrin), *DUOX2* (thyroid oxidase 2 or dual oxidase 2). The defects can lead to lower clinical expression of these genes with development

of goiters of small or medium sizes when the iodine nutritional environment is favorable, that is, in conditions of low dietary iodine [22]. This fact is observed even in individuals with a compound heterozygous or homozygous defect in certain genes, such as the *IYD* (iodotyrosine deiodinase) or *DEHAL1*. In contrast, the genetic defect in the heterozygous condition (one mutant allele) can lead to mild phenotypic manifestations, leading to a more severe clinical picture in the presence of iodine deficiency [22].

Linkage studies have been carried out to analyze the coinheritance of different genomic regions, as well as to assess the genetic mechanisms involved in this process. The genome-wide linkage analysis identified the candidate locus MNG1 (multinodular goiter 1) on chromosome 14q31 in a large Canadian family with 18 affected members [25]. This locus was confirmed in another German family with simple goiter that recurred after successive surgeries [26]. Both families showed the dominant pattern of inheritance with high penetrance. It is also worth mentioning a study carried out in an Italian family which detected a link between NNG and the X chromosome (autosomal dominant) and a novel NNG gene mapping to the MNG2 (Xp22) locus which awaits confirmation [27].

To identify other causal genes, a genome-wide linkage study has been performed in 18 families with goiter in Denmark, Germany, and Slovenia. In 20 % of the families investigated, four novel candidate loci were identified in different chromosomes (2p, 3p, 7q, and 8p) [28]. An individual contribution of the 3p locus was attributed to four families, whereas the other loci were attributed to one family each. This suggests a dominant pattern of inheritance in NNG [28]. In summary, we can state that the multiple risk factors mentioned above are likely to interact with or activate genetic susceptibility leading to the disorder. In contrast to sporadic NNG, which is caused by spontaneous recessive genomic variations, most cases of familial NNG present an autosomal dominant pattern of inheritance, indicating predominant genetic defects.

Physiopathology of NNG

The pathogenesis of goiter has been detailed elsewhere [1, 22]. An adapted summary of its main aspects is presented below.

Thyroid gland hyperplasia

Nodular goiters result from hyperplasia of follicular cells in one or several areas of the thyroid. The basic goitrogenic process is the development of new follicular cells forming new follicles or increasing the size of newly formed follicles. The driving force on the development of an NNG is

an abnormal potential for intrinsic growth in a small number of thyrocytes. Genetic, endogenous, and environmental factors can act on this basic process and accelerate growth. TSH is the most important stimulator of growth and thyroid function under physiological conditions *in vivo*.

Higher serum clearance of iodide

In chronic iodine deficiency, the thyroid maintains a constant concentration of iodine by increasing the clearance of plasma inorganic iodine. This phenomenon has been described by several authors [8]. The relationship is such that the product of thyroid clearance and iodine concentration is constant within the observed range of serum iodine concentrations. This product represents absolute iodine uptake, which is the mass of iodine available to the gland per unit of time. Despite the elevated clearance, absolute iodine uptake tends to be lower in iodine-deficient areas, indicating that the compensatory mechanism is neither perfect nor complete [8]. The increased iodine uptake reflects TSH stimulation and an intrinsic autoregulatory mechanism dependent on the intrathyroidal iodine concentration. The process of adjustment tends to decrease with time due to a continuous morphological deterioration of the thyroid, progressing from diffuse to multinodular hyperplasia when the goiter loses its adaptive efficiency.

Oxidative stress as a negative influence on thyroid hormone synthesis

In addition to being a substrate for hormone synthesis, H_2O_2 may be an important source of free radicals and reactive oxygen species. Since these elements may cause substantial damage and interfere with the normal cellular function, the epithelial cells of the thyroid seem to have an important defense mechanism against them in antioxidant enzymes, such as glutathione peroxidases, to counteract the potential damage mediated by free radicals. If this antioxidative defense is not effective, detectable damage (such as peroxidation) may occur in lipids, DNA, and thyrocyte proteins [21].

As iodine and H_2O_2 act as cosubstrates in thyroid hormone synthesis, changes in iodine concentrations are likely to affect the concentration of H_2O_2 . The generation of H_2O_2 , required for the incorporation of iodine, is also stimulated by TSH. Thus, low levels of iodine and increased thyroid function induce activation of H_2O_2 , which may result in DNA damage and somatic mutations [29]. Consequently, low iodine and high H_2O_2 intensify the antioxidative defense, which may be detected in cellular regulation of enzymes involved in the defense against oxidative stress.

In fact, studies in rodents have demonstrated a higher expression of antioxidant enzymes, especially SOD3

(superoxide dismutase 3), which act preferentially in the follicular lumen. These enzymes are likely to shield the thyroid from the toxic effects of H_2O_2 metabolites and thereby protect the gland from DNA damage and somatic mutations induced by these products [30]. In addition, oxidative stress and antioxidative defenses are increased in the presence of gland hyperplasia and involution [22].

Nodular transformation with progression to NNG

Most goiters become nodular with time. Along with the hyperplasia caused by the decrease in iodine, there is an increase in functional activity accompanied by significant increase in the number of thyroid cells. This increase in cell replication hinders the repair process of the cell, favoring an excess of mutations in the thyroid that randomly affects genes crucial for the normal physiology of the thyrocyte. Mutations that confer a growth advantage, particularly in the gene encoding the TSH receptor or $G_s\alpha$ protein, are the most likely initiators of focal growth [1].

The mechanism involved in the nodular transformation of a normal thyroid into a goiter may be hypothesized from epidemiological studies, animal models, and molecular data [1]. Initially, marginally low or deficient nutritional iodine and the concurrent action of goitrogens (see Tables 1, 2) cause diffuse thyroid hyperplasia. After this stage, increased cell proliferation, with possible DNA damage due to H_2O_2 action, cause mutational burden. Some of these spontaneous mutations induce constitutive (TSH-independent) activation of the cyclic AMP cascade, resulting in stimulation of growth and function. Finally, with the proliferation of cells in the thyroid, there is expression of growth-promoting factors, such as EGF (epidermal growth factor), IGF1 (insulin-like growth factor 1 or somatomedin C), FGF1 (fibroblast growth factor 1 [acidic]), and FGF2 (fibroblast growth factor 2 [basic]). As a result of the expression of these factors even at low concentration, the small clones with activating mutations continue to proliferate if capable of self-stimulation. They can then combine into small cell foci and develop as nodules. Nodular areas in the same gland exhibit different patterns of iodine uptake and cyclic AMP generation, confirming the presence of functional heterogeneity in NNGs [31] (Fig. 1).

In summary, the development of a nodule in an NNG is possibly the result of inherent and acquired heterogeneity in the proliferative and functional upregulation of thyroid follicular cells. These cells are intrinsically heterogeneous with regard to hormone production and proliferation in response to TSH stimulation. This is reflected in the different behaviors under intermediate levels of stimulation in which a subpopulation of cells outgrows others and expands into macroscopic nodules. In contrast, follicular cells acquiring activating somatic mutations in

cell proliferation pathways can expand clonally and also develop into a nodule. About 60–70 % of the thyroid nodules develop through the later mechanism and are monoclonal in origin, as a result of a neoplastic process with somatic mutations as the starting point, although the precise gene abnormalities are unknown. Somatic mutations leading to constitutive activation of TSH receptors are found in about 60 % of autonomously functioning nodules. In the remaining 40 % of the functioning nodules, no mutation is found in the TSH receptor, and the genetic mechanisms involved are poorly understood.

The pathogenesis of NNG encompasses processes of diffuse follicular hyperplasia, focal nodular proliferation, and eventual acquisition of functional autonomy. The development of an NNG is a result of a presumed inborn error of thyroid hormone synthesis associated with long-term exposure of the thyroid to proliferative stimuli such as iodine deficiency, goitrogens, or other factors. All the above result in insufficient thyroid hormone production and stimulate pituitary secretion of TSH.

Natural history

The natural history of the growth and function of the thyroid gland is variable and difficult to predict. In certain populations, the degree of spontaneous increase has been estimated at approximately 20 % per year, although this percentage may be much lower.

Individuals with NNG can present with hyperthyroidism or, less often, hypothyroidism. Hyperfunction often develops insidiously, in contrast to what is observed in Graves' disease. It generally starts with prolonged subclinical hyperthyroidism, characterized by low serum TSH and normal serum free T₄, T₃, and total T₄ levels. This is a result of the growth of the goiter and is associated with the increase in the volume of cells producing thyroid hormones autonomously. The following observations support the concept of increasing thyroid nodularity and autonomy of thyroid function related to the increase in thyroid volume during the natural history of NNG: (a) the volume of the thyroid is greater in older subjects; (b) the longer the individual has a goiter, the larger the size of that goiter; and (c) the larger the size of the goiter, the lower the serum TSH concentration [32].

The rates of progression from a simple to a toxic NNG and the time that this takes to occur are poorly understood. A population study carried out in an iodine-deficient area showed that nodular autonomy increased with age and affected 15 % of the elderly [33]. According to some longitudinal studies, hyperthyroidism manifests in 9–10 % of the individuals with nodular goiter within a period of 7–12 years [34, 35]. In a few cases, the autonomy of some thyroid nodules may regress.

In contrast, the development of hypothyroidism in an individual with NNG is rare and often associated with coexisting autoimmune thyroiditis.

Diagnostic approach in NNG and NDG

Clinical evaluation

Clinical signs and symptoms associated with NNG are shown in Table 3, and the diagnostic aspects related to the disease are depicted in Table 4.

The occurrence of symptoms of compression of cervical structures (trachea, great vessels, and recurrent laryngeal nerve) is almost always an indication that a long-term NNG has partially migrated to the retrosternal and upper mediastinal regions (Table 2). This occurs more frequently in individuals older than 40–50 years, possibly due to changes in the cervical spine (decreased vertebral spaces, vertebral listhesis, and, possibly, cervical osteoporosis) projecting parts of the NNG toward the retrosternal region. When this occurs, the goiter may compress the jugular and subclavian veins in the area around the superior vena cava.

In the Pemberton maneuver, dislocation of a goiter into the upper thoracic aperture obtained by extension of the arms above the head may cause respiratory difficulty, distension of the neck veins, facial congestion, and stridor due to increased pressure on the trachea (Pemberton's sign) [36]. This phenomenon seems to be uncommon. In the medical literature, description of tracheal compression triggering cough or stridor is reported in about 18 % of the individuals with large NNGs [37]. Thoracic inlet imaging is often requested in patients with retrosternal goiter after a finding of extra-thoracic airway obstruction

Table 3 Clinical signs and symptoms that may be associated with nodular goiter

-
- Slow and progressive growth of the thyroid
 - Uni- or multinodularity on examination
 - Cosmetic complaints
 - Tracheal compression or deviation, upper airway obstruction, dyspnea
 - Occasional cough or dysphagia
 - Acute pain or cervical increase secondary to bleeding
 - Superior vena cava obstruction syndrome
 - Pemberton's sign: upper thoracic inlet obstruction during extension of the arms above the head
 - Gradual development of hyperthyroidism
 - Iodine-induced thyrotoxicosis
 - Recurrent nerve palsy (rare)
 - Phrenic nerve palsy (rare)
 - Horner's syndrome (rare)
-

Table 4 Characteristics of benign nodular goiter

- Multinodularity on physical examination
- Tracheal deviation, asymmetry
- Absence of cervical adenopathy
- Normal or decreased TSH, normal or increased free T4, high thyroglobulin levels
- Normal calcitonin
- Negative thyroid autoantibodies in ~90 % of cases
- Scintigraphy showing “hot” and “cold” areas
- Finding of nodularity on ultrasonography: cysts and calcifications are common
- Heterogeneous mass demonstrated by computed tomography and magnetic resonance
- Respiratory function assessment may demonstrate impaired inspiratory capacity
- Benign cytology in dominant nodules at FNA biopsy

in a flow-volume loop spirometry or a mediastinal shadow causing tracheal deviation on chest X-ray. Very often, computed tomography (CT) images show a decrease in more than 50 % in the tracheal segment due to compression in the absence of complaints or discomfort. Other findings may include esophageal involvement, vocal cord paralysis, phrenic nerve palsy, and Horner’s syndrome due to damage to the cervical sympathetic chain [38]. In rare instances when an NNG becomes malignant, symptoms of compression, shortness of breath, and a persistent cough that worsens in the recumbent position should alert for the possibility of malignancy.

It is important to emphasize that hyperthyroidism may develop as an NNG increases progressively in size. This dysfunction is almost always caused by excessive iodine exposure, such as excessive dietary iodine, accidental ingestion of drugs containing iodine, use of povidone as an anti-bactericidal agent, dermal application of iodine, and use of cosmetics containing red dye (which is iodized) [39].

In hyperthyroidism secondary to excessive dietary iodine intake, thyrotoxicosis develops acutely. In contrast, the development is often insidious in long-term NNGs, in which hyperthyroidism occurs due to the development of autonomous thyroid nodules [40]. Since the occurrence of hyperthyroidism in this setting usually affects the elderly, cardiac arrhythmia is commonly the initial presentation, particularly in the form of atrial fibrillation. In some individuals, scintigraphy may show a dominant hyperfunctioning (autonomous or “hot”) nodule. In others, heterogeneous distribution of the radionuclide is more commonly observed. Thyroid scintigraphy is usually the method of choice to confirm the diagnosis of hyperthyroidism in NNG caused by excess iodine, in which a very low isotope uptake by the thyroid is demonstrated.

During the clinical assessment, affected individuals should be asked about the familial occurrence of benign or malignant thyroid disease.

The problem of malignancy in NNGs

The risk of malignancy in a nodule within a multinodular goiter has not been completely elucidated. Recent guidelines for the management of thyroid nodules recommend the screening to be based on the assumption that the risk of malignancy in a nodule within a multinodular goiter is comparable to that in single nodules [7]. However, evidence from individual studies shows that the reported prevalence of thyroid malignancy in patients with NNG varies between 4 and 17 %. These high prevalences may be due to inclusion of incidental papillary cancers detected on surgical specimens. [41–43], in contrast to an estimated prevalence of 5 % in those with a single nodule [44]. This shows a great discrepancy between these rates, which may be due to the selection process of the patients. The histological criteria for malignancy are also variable, and it is notably difficult to predict with any degree of certainty the growth potential of a particular lesion. These carcinomas vary widely in size and are typically papillary and well differentiated; therefore, their prognosis is remarkably good, with a reported survival of 95 % in 30 years [45]. These aspects should be taken into account when conservative therapy is under consideration.

A family history of NNG suggests the occurrence of a benign disease but does not exclude the possibility of malignancy. It is important to obtain information on the occurrence of familial papillary carcinoma, which would indicate the need for better and more comprehensive laboratory and cytological investigation.

The risk of malignancy in NNGs is higher in young individuals and in the elderly [2]. A history of ionizing radiation in the cervical region during childhood should alert to the possibility of a papillary carcinoma. Likewise, a more evident growth of part of an NNG should alert to the possibility of malignancy. In contrast, a rapid increase in volume in a cyst, or the presence of a hemorrhagic area can be easily diagnosed by ultrasonography.

Thyroidectomy should be considered irrespective of a benign cytology in cases with a high clinical suspicion of malignancy evidenced by rapid tumor growth, unusual nodular induration, adherence to adjacent structures, vocal cord paralysis at laryngoscopy, and regional lymphadenopathy [2].

Laboratory assessment

Initial laboratory tests most often requested are serum TSH and free T4. This practice is consistent with the

Table 5 Routine biochemical measurements used by ATA, ETA, and LATS members in the complementary diagnostic assessment of individuals with simple NNG with no suspicion of malignancy

Measurements	Nontoxic nodular goiter		
	ATA (<i>n</i> = 140)	ETA (<i>n</i> = 120)	LATS (<i>n</i> = 148)
TSH	100	100	96
Total T4	21	17	20
Total T3	23	23	22
Free T4	54	74	63
AntiTPO	61	65	76
AntiTG	34	8	32
Calcitonin	4	32	5
Thyroglobulin	2	8	6

Numbers percentage of members from the organizations. *ATA* American Thyroid Association, *ETA* European Thyroid Association, *LATS* Latin American Thyroid Association, *antiTPO* antithyroperoxidase antibodies, *antiTG* antithyroglobulin antibodies. Adapted from [62]

recommendations and preferences of experts in the United States and Europe (Table 5).

Measurement of these hormones assesses the function in an NNG and evaluates the possibility of subclinical hyperthyroidism. The presence of positive anti-thyroid peroxidase (antiTPO) and antiTG antibodies reflects concomitant autoimmune thyroid disease [2]. Levels of TG are consistently high and correlate with the size of the NNG, but add little information from a diagnostic point of view. It is beyond the scope of this review to discuss the role of measurement of calcitonin to rule out medullary thyroid cancer (MTC) in an NNG. International studies offer no consensus on this issue and in our opinion this investigation is not justified due to the large number of individuals with NNG, and considering the high cost of the test and the low prevalence of MTC (0.4–1.4 %) [46].

Imaging investigation

Inspection and palpation of the thyroid by the clinician are the initial steps in the investigation of an NNG. This should include a careful description of the features of the nodules, position, texture, signs of calcification, areas with possible recent and painful hemorrhage, and signs of venous compression or tracheal compression in the supine position. However, clinical evaluation of NNGs is rather inaccurate, particularly when the NNG extends to the retrosternal and mediastinal regions. Therefore, this evaluation should be followed by an imaging assessment to correlate the clinical information with data obtained from imaging tests.

Ultrasonography

Routine assessment of the thyroid with ultrasonography (in association with the analysis of the vascularization with color Doppler) had a great impact in endocrinology practice [47]. This methodology is widely disseminated due to the greater availability of high-resolution equipment, relatively affordable devices, little or no discomfort for the patient, and the absence of ionizing radiation. Additionally, it provides valuable information about the nodules (presence of halo, microcalcifications, echogenicity, location, volume), and offers precise directions for sample collection by fine-needle aspiration (FNA). It is then not a surprise that 80 % of the endocrinologists choose this resource for initial imaging assessment in patients with NNG.

However, it is necessary to stress that ultrasonography is a procedure dependent on the observer, in which professionals with greater experience and larger number of patients will have a better diagnostic ability when compared with inexperienced ones. Due to that, divergent reports issued by different diagnostic centers are common. Also, experts disagree on the role of ultrasound to identify malignant features in thyroid nodules. In our view, the use of already established criteria such as the absence of a halo, undefined borders, presence of microcalcifications, marked hypoechogenicity, and central vascular flow at Doppler evaluation may indicate the need to complement the assessment with a cytological analysis [47]. Reported mean specificities for predicting malignancy are 67 % for marked hypoechogenicity, 70 % for microcalcifications (small, intranodular, punctate, hyperechoic spots with scanty or no posterior acoustic shadowing), 70 % for irregular or microlobulated margins, and about 80 % for chaotic arrangement or intranodular vascular images [48, 49]. The value of these features for predicting cancer is partially blunted by low sensitivity attributed to these findings, but no ultrasonographic feature alone is entirely predictive of malignancy [50]. Moreover, certain ultrasonographic features may be suggestive at best, but are not a proof of malignancy.

Ultrasonography allows a precise assessment of the volume of an NNG, providing the gland does not extend behind the sternum or to the upper mediastinum. If carried out by the same investigator, volumetric variations can be detected, which may lead to medical or surgical intervention.

There is a lack of clinical criteria for malignancy in most nonpalpable nodules [51, 52]. Hence, it is essential to determine which thyroid lesions have a high potential for malignancy based on ultrasonographic features. The prevalence of cancer detected by ultrasonographic characteristics suggestive of malignancy in nonpalpable thyroid nodules is similar to that in palpable nodules [49, 53]. The occurrence

of malignancy is not less frequent in nodules smaller than 10 mm in diameter than in those larger than that. Therefore, an arbitrary diameter cutoff for cancer risk is not justified, and suspicious lesions smaller than 10 mm should be assessed with FNA biopsy when warranted [50]. Since an aggressive disease course is rare in incidentally discovered microcarcinomas [54], incidental thyroid lesions with a diameter of about 5 mm are usually followed up with ultrasonography [55]. There is no evidence that surgical intervention in nonpalpable nodules is associated with health improvement in affected individuals.

In comparative studies assessing medium to large NNGs, the accuracy of ultrasonography is lower than that of CT, particularly when part of the thyroid tissue is retrosternal.

Scintigraphy

Thyroid imaging with radionuclides (radioiodine and technetium) has been performed for many years. However, as a method to assess volumetric and morphological features, scintigraphy leaves much to be desired when compared with ultrasonography.

Scintigraphy is useful to evaluate the functional status of the nodules in an NNG, identifying nodules with high uptake of radionuclide along with others with scarce radioactive concentration. The use of ^{99}Tc (technetium) as a tracer may be associated with a false-positive uptake in 3–8 % of the nodules, which does not occur with radioiodine [56]. Evidently, the presence of subclinical or clinical hyperthyroidism should raise the suspicion of hyperfunctioning autonomous nodule in the context of multinodularity and, in this situation, scintigraphy may be useful for diagnostic purposes.

Computed tomography, magnetic resonance imaging (MRI), and positron-emission tomography (PET)

These methods provide high-resolution, three-dimensional visualization of the thyroid gland. However, due to a high dose of ionizing radiation, there is no advantage in performing CT routinely in NNGs, except in glands with retrosternal or upper mediastinal extension, which are not adequately visualized by ultrasonography. Another role of CT is on the follow-up after treatment with radioiodine preceded by recombinant human TSH (rhTSH) to determine with accuracy the volumetric reduction of retrosternal NNGs [56]. Use of iodinated contrast during CT should be avoided due to the risk of the Jod-Basedow phenomenon. Regarding MRI, there is a lack of comparative and accurate studies with reliable conclusions about the use of this method when compared with CT. In the routine assessment of NNGs in hospitals and clinics, MRI has a limited role for the assessment of nodular volume or characteristics.

Positron-emission tomography with radioactive fluoro-deoxyglucose (FDG-PET) should also not be used for evaluation of an NNG, although some reports have shown it to be useful in solitary solid nodules, in which this method may indicate the possibility of malignancy [57].

Fine-needle aspiration biopsy

This procedure provides direct and accurate information on the cytology of a single nodule. In individuals with NNG, malignant nodules are mostly indistinguishable from benign ones from a clinical point of view. Therefore, a consensus based on several recommendations of associations from Europe and North America is to perform an FNA biopsy of at least two nodules according to previous ultrasonographic findings associated with malignancy [58]. The coexistence of two or more suspicious ultrasonographic findings greatly increases the possibility of thyroid cancer. Also, in the presence of suspicious cervical lymphadenopathy, FNA biopsy of both the lymph node and the suspicious nodule(s) is essential.

Therapeutic options

Once the diagnosis of NDG or NNG is established, the following management goals should be considered:

- (a) Correct the underlying thyroid dysfunction, if present;
- (b) Verify if the goiter is growing or causing obstructive symptoms;
- (c) Exclude malignancy if one or more nodules are suspicious;
- (d) Determine whether the goiter requires therapy and if so, weight the benefits and risks of medical and surgical interventions and reach a decision with the patient about what type of treatment should be administered.

The decision of how to treat an individual with benign nontoxic goiter can be difficult, especially since there is no uniform association between the size of the goiter and the presence of symptoms.

Nontoxic diffuse goiter

Treatment of patients with NDG, when considered to be indicated, is medical rather than surgical. It is unclear whether or not the early treatment of NDG can inhibit the development of nodular goiter [59].

As discussed previously, NDG results from a combination of genetic factors and environmental conditions. Prolonged stimulation by TSH, which frequently results from iodine deficiency, is considered an important factor for thyroid enlargement. Therefore, iodine supplementation in this

situation would seem an adequate approach. In fact, supplementation with iodine 400 µg/day for 8–12 months has been demonstrated to be as effective as suppressive therapy with LT4 150 µg/day [60]. However, except in a few European countries, iodine is no longer used for the treatment of benign nontoxic goiter.

In contrast, a beneficial effect of LT4 has been shown in NDG [61]. If suppressive treatment is considered, then administration of thyroid hormone in enough doses to inhibit or reduce TSH secretion may be used. However, it should be considered that the volume of the goiter returns to pretreatment size after LT4 withdrawal.

Nontoxic nodular goiter

There is no consensus on the ideal treatment for NNG, due in part to the variable natural history associated with the disease. In some individuals, the goiter increases gradually in size over time and is accompanied by the development of multiple nodules, compressive symptoms, and cosmetic concerns [2]. However, in about 20 % of the women and 5 % of the men, the goiter stabilizes or regresses spontaneously over time [6].

Options for management of NNG include:

- Clinical observation for asymptomatic patients;
- Thyroid hormone suppressive therapy;
- Radioiodine therapy alone or preceded by rhTSH; and
- Surgery.

Among these options, treatment is chosen individually for each patient in view of the risks, benefits, and availability of the various techniques, experience of the treating physician, and patient's personal preference.

Clinical observation

Observation alone may be appropriate in individuals with normal thyroid function, who are asymptomatic, and without cosmetic concerns after the diagnosis of benign nontoxic goiter is established by clinical evaluation, and laboratory and thyroid imaging tests. In general, individuals with asymptomatic NNG may be followed yearly with careful palpation of the thyroid and serum TSH measurement. If the findings on palpation are unclear, ultrasonography may be indicated. Nodules that appear to be growing may require FNAB. Neck CT or MRI may be necessary if the goiter extends to the retrosternal area.

Levothyroxine suppressive therapy

Management of NNG with thyroid hormone is controversial. According to surveys, suppressive therapy with LT4

is used extensively in the United States, Europe, and Latin America (Table 6) [62]. The efficacy of LT4 depends on the degree of TSH suppression. Since most patients with NNG have normal serum TSH concentration, thyroid enlargement in these patients is probably caused by several growth factors (including TSH) acting over time on thyroid follicular cells with different synthetic and growth potentials.

Potential adverse effects are more noticeable in older patients, who comprise the larger proportion of individuals with NNG. In addition, up to 22 % of the affected individuals may have important functional autonomy rendering LT4 therapy more problematic due to an increased risk of bone loss and atrial fibrillation.

When suppressive therapy with LT4 is initiated, long-term treatment is required. Thyroid hormone is presumed to reduce goiter size in some individuals by reducing TSH secretion, particularly in those from regions with borderline or low iodine intake. Since any reduction in NNG size during therapy is lost once LT4 is withdrawn, there is a possibility of recurrence of nodular and/or glandular growth [63]. However, the maintenance of serum TSH level in the lower end of or just below the normal range, rather than below 0.01 µU/mL, appears to be efficacious. Pending further study, this strategy may relieve the adverse effects of LT4 therapy [64].

A potential benefit of thyroid hormone therapy is a reduction in the risk of thyroid oncogenesis [65]. However, this hypothesis is still speculative, and additional studies are still required.

Overall, since LT4 suppressive therapy is associated with a reduction in the volume of NNGs in only about 30 % of the patients, the choice of this treatment modality has declined [2].

Table 6 Comparison of treatment preferences among members of the ATA, ETA, and LATS thyroid associations based in an index case with simple NNG with no suspicion of malignancy

Treatment	Nontoxic nodular goiter		
	ATA (n = 140)	ETA (n = 120)	LATS (n = 148)
No treatment	36	28	39
LT4	56	52	21
¹³¹ I ^a	1	6	7
Surgery	6	10	28

Numbers percentage of members from the organizations, *ATA* American Thyroid Association, *ETA* European Thyroid Association, *LATS* Latin American Thyroid Association, *LT4* levothyroxine, ¹³¹I radioiodine. Adapted from [62]

^a Includes both isolated radioiodine use and radioiodine associated with rhTSH

Surgery

The optimal surgical procedure for NNG is still debated. When surgery is the treatment of choice, total thyroidectomy is recommended over subtotal thyroidectomy for most patients with large obstructive goiters, cosmetic complaints, or retrosternal NNGs [66, 67]. Subtotal thyroidectomy is associated with higher recurrence rates than total thyroidectomy, requiring reintervention in 2.5–42 % of the patients over the long term. Also, 3.5 % of the patients with NNG treated with subtotal thyroidectomy require a second procedure to remove all thyroid tissue due to incidental thyroid cancer [68]. Both total and subtotal techniques have similar rates of permanent complications such as hypoparathyroidism and vocal palsy, but total thyroidectomy is preferred since the risk of these complications increases with reintervention [69]. In patients with unilateral NNG, some authors recommend unilateral thyroidectomy based on a low rate of recurrence (2 %) and high rate of maintenance of euthyroidism (73 %) [70]. Postsurgical recurrence rates are directly proportional to the volume of the remaining thyroid tissue.

Usually, intrathoracic goiters can be approached through a cervical incision, although it has been reported that 10–30 % of the cases require sternotomy or thoracotomy [71]. The complication rates are higher in individuals with NNG with retrosternal extension when compared with individuals whose goiters are located exclusively in the cervical area [72].

Total thyroidectomy should be followed by LT4 replacement therapy at a dose of 1.4–2.2 µg/kg/day [73]. Appropriate adjustments should be made following routine recommendations for patients with hypothyroidism. If partial thyroidectomy is chosen, LT4 should only be started after the development of hypothyroidism, and not as a preventive measure against goiter recurrence since randomized trials have failed to confirm such benefit [74].

Since the incidence of NNG increases with age, the benefits of surgery must be carefully weighed against possible risks associated with the procedure, especially in the elderly who may present substantial comorbidity. Alternatively, radioiodine preceded by rhTSH stimulation may be an option, as discussed below in this section.

Radioiodine therapy

Radioiodine therapy is recommended for NNG patients with contraindication for or refusal of surgery. This treatment modality in nodular goiter has increased in recent years and results in a considerable reduction in thyroid volume, with rates of 30–40 % in the first year, and 50–60 % on the fourth year. Most individuals report improvement of obstructive symptoms [75], with reports of a single dose

administered orally restoring euthyroidism over a period of 2–4 months [56].

The larger the volume of the gland and the lower the radioiodine uptake, the higher should be the radioiodine activity to be administered. Subsequent studies have unanimously confirmed this concept [76]. In very large goiters (>100 cm³), the decrease in glandular volume is lower (around 35 %) despite the use of equivalent doses of radioiodine [76].

Recombinant human TSH-stimulated radioiodine therapy

A frequent limitation of the treatment with radioiodine is the low accumulation level of the isotope in inactive and partially suppressed paranodular areas in NNGs. This problem may be solved by increasing the radioiodine uptake (RAIU) in such nodular goiters [77]. For this purpose, recent studies using rhTSH in preparation for radioiodine therapy have shown good results [78].

Therapy with rhTSH in combination with radioiodine is well tolerated and is associated with similar side effects as those observed when the radioiodine is administered alone. However, in the first 48 h after radioiodine therapy, there is a brief elevation in thyroid hormone levels causing transient mild thyrotoxicosis followed by possible hypothyroidism within the first month after the treatment [79, 80]. Other observed acute adverse effects are painful transient thyroiditis, thyroid swelling, tracheal compression, and, frequently, cardiac symptoms. These conditions may be minimized by the use of glucocorticoids and β-blockers [81].

Several recent articles [77] have indicated that these adverse effects are probably dose-dependent and are negligible with lower rhTSH doses. According to Bonnema et al. [77], the optimal dose of rhTSH to improve radioiodine therapy is most likely in the range of 0.03–0.1 mg. Significant improvement in RAIU by the thyroid is obtained in this dose range, with minimization of the risk of glandular swelling and temporary thyrotoxicosis.

An interesting, diverse approach to increasing the thyroid RAIU is to stimulate the secretion of endogenous TSH with methimazole [82]. However, whether a marginal hypothyroid state obtained with methimazole is as effective as rhTSH to increase the thyroid RAIU and to augment the reduction in goiter size after radioiodine therapy remains to be clarified with controlled trials.

Table 7 shows different situations associated with a multinodular goiter and the recommended therapy (surgery or radioiodine) for each of them.

In summary, individuals with NDG should receive clinical rather than surgical treatment. NNG is a highly prevalent disease, even in regions without iodine deficiency. Many individuals are asymptomatic, and when symptoms

Table 7 Situations in which thyroid surgery or radioiodine may be preferred in a patient with multinodular goiter

	Radioiodine	Surgery
Small or moderate goiter volume	++ ^a	+
Previous thyroidectomy	++	+
Coexisting hyperthyroidism due to nodular autonomy	++	+
Severe comorbidity	++	–
Suspicion of thyroid malignancy	–	++
Very large (>100 cm ³) benign goiter without obstructive symptoms	+ ^b	++
Severe tracheal compression	+	++
Need of rapid relief of goiter symptoms	+	++
Insufficient response to previous radioiodine therapy	+ ^c	++
Retrosternal goiter or intrathoracic extension	++ ^d	+

Adapted with modifications from [76]

++ First choice, + second choice, – no option

^a Observation may be considered in euthyroid patients with asymptomatic diffuse or nodular benign goiters

^b RhTSH-stimulated radioiodine therapy may be considered

^c A second course of radioiodine therapy may sometimes lead to a satisfactory result

^d Many retrosternal or intrathoracic goiters are large, which favors surgery

are present, the most common clinical manifestations result from local compressive effects. Affected individuals should undergo a detailed investigation in order to rule out malignancy. When that is achieved, therapeutic modalities should be individualized weighing the risk/benefit of each option and discussed with patients. Total thyroidectomy is the first treatment option, followed by the use of radioiodine alone or after rhTSH pretreatment to increase the efficacy of the radioiodine treatment.

The efficacy of thyroid hormone suppressive therapy in NNG is questionable, and its use has declined due to concerns about potential long-term side effects associated with the induction of subclinical hyperthyroidism and the fact that most goiters increase in volume again after discontinuation of the thyroid hormone. Each of these therapeutic options has advantages and disadvantages, with acute and long-term side effects.

Final remarks

Apart from local signs and possible progression with time to subclinical or overt thyrotoxicosis, there is no extrathyroidal involvement in NDG and NNG like the occurrence of ophthalmopathy in Graves' disease. The local signs associated with NDG and NNG are those of any other space-occupying tumors in the area of the upper thoracic outlet.

Sporadic NNG is not usually considered a predisposition to thyroid cancer, although it may hinder the detection of malignancy. Accelerated growth of a solitary nodule within any goiter is the single most alarming sign, although even entirely benign nodules may grow rapidly.

Neck palpation is well-known to be an imprecise assessment to evaluate the morphology and determine the size of the thyroid. In contrast, high-resolution ultrasonography is the most sensitive method to detect thyroid nodules. Both CT and MRI have no advantage over ultrasonography for visualization of the intrathyroidal structure and are not recommended for routine evaluations. Their major strength is their ability to assess patients with large goiters with suspected retrosternal extension, or obstructive or compressive symptoms. Scintigraphy with iodine isotopes may be unsuitable because of the frequent lack of uptake in different areas of large nodular goiters.

The goals of management are to correct the underlying thyroid dysfunction, if present, and to decrease the size or prevent further growth of goiter. In the absence of adequate randomized trial data, management decisions should be individualized based upon patient characteristics. Many patients with benign but large goiters may experience clinical symptoms of pressure, such as dysphagia, choking sensation, or airway obstruction. These patients often require surgery to mitigate their symptoms. In the absence of malignancy, asymptomatic patients may be observed. The effectiveness of treatment with LT4 to reduce the volume of the goiter remains controversial. Radioactive iodine alone or with rhTSH is safe and effective and may be a reasonable therapeutic option.

What is the optimal treatment for benign NDG and NNG goiters? Once the diagnosis and indication for treatment of nontoxic goiter has been made, the treating physician and patient should discuss each of the treatment options, including the logistics, benefits, expected speed of recovery, drawbacks, side effects, costs and then decide on the

best treatment modality for that particular patient, taking into account the patient's age and comorbidities.

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Compliance with ethical standards

Conflict of interest The author has no potential conflict of interest to declare.

Ethical approval No studies with humans or animals were performed by the author in the preparation of this review.

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